



آمار و احتمال مهندسی

Engineering Probability & Statistics

Lecture Notes ▪ جزوه درسی

Module 13

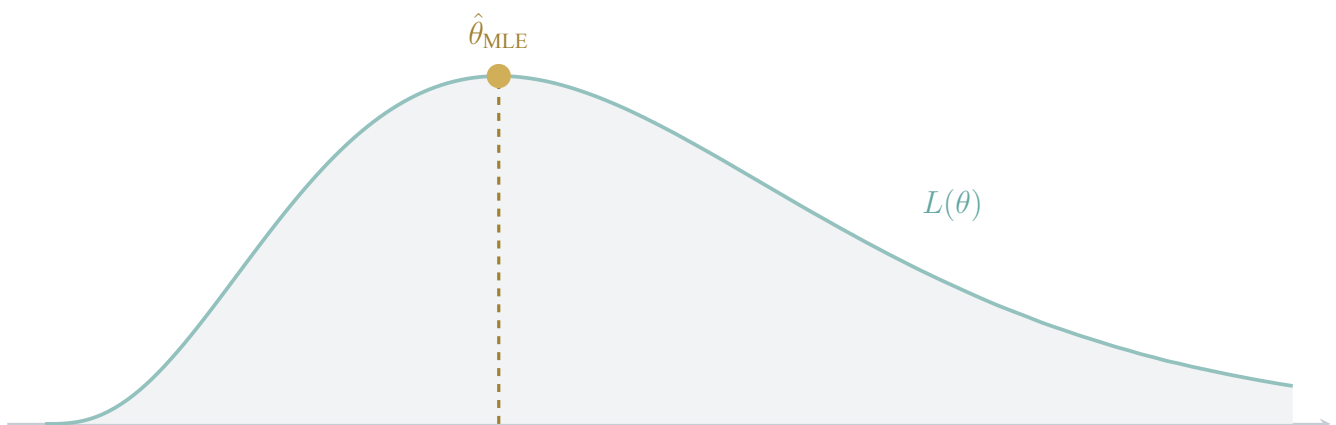
Parameter Estimation

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Learning Objectives

After completing this module, students will be able to:

1. Distinguish between estimator (rule) and estimate (number)
2. Evaluate estimator properties: bias, consistency, efficiency, MSE
3. Apply the Method of Moments
4. Apply Maximum Likelihood Estimation (MLE)
5. Compare estimators using MSE and bias-variance tradeoff

13.1 Introduction

We observe data $X_1, \dots, X_n$ from a distribution with unknown parameter θ . **Parameter estimation** gives us a method to infer θ from data.

Engineering examples:

- ▶ Estimate noise variance σ^2 from recorded samples
- ▶ Estimate failure rate λ from component lifetimes
- ▶ Estimate channel gain from received signal strength
- ▶ Estimate mean signal level from measurements

13.2 Estimator vs. Estimate

■ Definitions

Term	Type	Definition	Example
Estimator $\hat{\theta}$	Random variable (function of data)	The RULE applied to data	$\hat{\theta} = \bar{X} = \frac{1}{n} \sum X_i$
Estimate	Number	The VALUE from a specific sample	$\hat{\theta} = 3.47$ (from one dataset)

■ Key Insight

The **estimator** is a random variable — it has a sampling distribution.

The **estimate** is one realization of that random variable.

Analogy: “Take the average” is the estimator (rule). “The average was 3.47” is the estimate (result).

13.3 Properties of Estimators

■ Bias

$$B(\hat{\theta}) = E[\hat{\theta}] - \theta$$

- **Unbiased:** $B(\hat{\theta}) = 0$, i.e., $E[\hat{\theta}] = \theta$ (on average, correct)
- **Biased:** $E[\hat{\theta}] \neq \theta$ (systematic error)

Examples:

- $\bar{X}$ is unbiased for μ : $E[\bar{X}] = \mu$ ✓
- S^2 (with $n - 1$) is unbiased for σ^2 : $E[S^2] = \sigma^2$ ✓
- S (sample std dev) is biased for σ : $E[S] \neq \sigma$ (slight downward bias)

■ Consistency

$\hat{\theta}_n$ is **consistent** if $\hat{\theta}_n \rightarrow \theta$ in probability as $n \rightarrow \infty$.

Practical meaning: with enough data, the estimator gets arbitrarily close to the truth.

Sufficient condition: If bias $\rightarrow 0$ and variance $\rightarrow 0$ as $n \rightarrow \infty$, then consistent.

■ Efficiency

Among all unbiased estimators, the **efficient** one has the smallest variance.

The **Cramér-Rao Lower Bound** (CRLB) gives the minimum possible variance:

$$\text{Var}(\hat{\theta}) \geq \frac{1}{nI(\theta)}$$

where $I(\theta)$ = Fisher information = $-E \left[\frac{\partial^2 \ln f(X; \theta)}{\partial \theta^2} \right]$.

An estimator achieving the CRLB is called **efficient** or **minimum variance unbiased (MVU)**.

■ Mean Squared Error (MSE)

$$\text{MSE}(\hat{\theta}) = E[(\hat{\theta} - \theta)^2] = \text{Var}(\hat{\theta}) + [B(\hat{\theta})]^2$$

$$\text{MSE} = \text{Variance} + \text{Bias}^2$$

This is the **bias-variance tradeoff**: sometimes a slightly biased estimator with much lower variance has smaller MSE than an unbiased one.

13.4 Method of Moments (MoM)

■ Principle

Set population moments equal to sample moments, then solve for parameters.

- ▶ k – th population moment: $\mu'^k = E[X^k]$
- ▶ k – th sample moment: $m'^k = \frac{1}{n} \sum X_i^k$

■ Procedure

1. Express parameters in terms of population moments
2. Replace population moments with sample moments
3. Solve for parameter estimates

Example: Exponential Distribution

Data from $\text{Exp}(\lambda)$. Population moment: $E[X] = \frac{1}{\lambda}$.

Set equal: $\frac{1}{n} \sum X_i = \frac{1}{\lambda} \rightarrow \hat{\lambda}_{\text{MoM}} = \frac{1}{\bar{X}}$

Example: Gaussian Distribution

Data from $N(\mu, \sigma^2)$:

- ▶ 1st moment: $\hat{\mu} = \bar{X}$
- ▶ 2nd central moment: $\hat{\sigma}^2 = \frac{1}{n} \sum (X_i - \bar{X})^2$ (note: biased, uses n not $n - 1$)

■ Advantages/Disadvantages

- ✓ Simple, always applicable, closed-form solutions often available
- ✗ Not always efficient, can produce biased estimates, may give invalid parameter values

13.5 Maximum Likelihood Estimation (MLE)

■ Principle

Choose the parameter value that makes the observed data **most probable**.

■ Likelihood Function

Given data $x_1, \dots, x_n$ from $f(x; \theta)$:

$$L(\theta) = \prod_{i=1}^n f(x_i; \theta)$$

■ Log-Likelihood (usually easier)

$$\ell(\theta) = \ln L(\theta) = \sum_{i=1}^n \ln f(x_i; \theta)$$

■ MLE Procedure

1. Write the log-likelihood $\ell(\theta)$
2. Differentiate: $\frac{d\ell}{d\theta} = 0$
3. Solve for $\hat{\theta}_{\text{MLE}}$
4. Verify it's a maximum $\left(\frac{d^2\ell}{d\theta^2} < 0\right)$

■ Properties of MLE

1. **Consistent** (converges to true value)
2. **Asymptotically unbiased** (bias $\rightarrow 0$ as $n \rightarrow \infty$)
3. **Asymptotically efficient** (achieves CRLB for large n)
4. **Invariant:** If $\hat{\theta}$ is MLE of θ , then $g(\hat{\theta})$ is MLE of $g(\theta)$

Example: MLE for Gaussian Mean

Data $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ with σ^2 known.

$$\ell(\mu) = -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum (x_i - \mu)^2$$

$$\frac{d\ell}{d\mu} = \frac{1}{\sigma^2} \sum (x_i - \mu) = 0 \rightarrow \hat{\mu}_{\text{MLE}} = \bar{X}$$

Example: MLE for Exponential Rate

Data $X_1, \dots, X_n \sim \text{Exp}(\lambda)$.

$$\ell(\lambda) = n \cdot \ln(\lambda) - \lambda \cdot \sum x_i$$

$$\frac{d\ell}{d\lambda} = \frac{n}{\lambda} - \sum x_i = 0 \rightarrow \hat{\lambda}_{\text{MLE}} = \frac{n}{\sum x_i} = \frac{1}{\bar{X}}$$

Example: MLE for Gaussian Variance

Data $X_1, \dots, X_n \sim N(\mu, \sigma^2)$, μ known.

$$\ell(\sigma^2) = -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum (x_i - \mu)^2$$

$$\frac{d\ell}{d(\sigma^2)} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum (x_i - \mu)^2 = 0$$

$$\hat{\sigma}_{\text{MLE}}^2 = \frac{1}{n} \sum (x_i - \mu)^2$$

Note: With μ unknown, $\hat{\sigma}_{\text{MLE}}^2 = \frac{1}{n} \sum (x_i - \bar{x})^2$ — biased! (Uses n , not $n - 1$)

13.6 Engineering Examples

■ Estimating Noise Variance

Problem: Measure n samples of noise. Estimate noise power σ^2 .

Data: $x_1, \dots, x_n$ (noise voltage samples, assumed zero-mean)

MLE: $\hat{\sigma}^2 = \frac{1}{n} \sum x_i^2$ (biased by factor $\frac{n-1}{n}$)

Unbiased: $S^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2$ or if $\mu = 0$ known: $\frac{1}{n} \sum x_i^2$ is actually unbiased for $E[X^2] = \sigma^2$

■ Estimating Failure Rate

Problem: 20 components tested until failure. Lifetimes: $t_1, \dots, t_{20}$.

Model: $T \sim \text{Exp}(\lambda)$, parameter λ (failure rate).

MLE: $\hat{\lambda} = \frac{20}{\sum t_i} = \frac{1}{\bar{T}}$

If $\bar{T} = 500$ hours: $\hat{\lambda} = \frac{1}{500} = 0.002$ failures/hour.

■ Estimating Channel Parameter

Problem: Rayleigh fading channel. Observed power samples: $p_1, \dots, p_n$.

Model: Power $P \sim \text{Exp}\left(\frac{1}{\Omega}\right)$, where $\Omega = E[P]$ = average power.

MLE: $\hat{\Omega} = \bar{P} = \frac{1}{n} \sum p_i$

13.7 Comparing Estimators

■ Bias-Variance Tradeoff

Two estimators for σ^2 :

- $\hat{\theta}_1 = S^2$ (unbiased, variance = $\frac{2\sigma^4}{n-1}$)
- $\hat{\theta}_2 = \frac{1}{n} \sum (X_i - \bar{X})^2$ (biased by $-\frac{\sigma^2}{n}$, variance = $\frac{2\sigma^4(n-1)}{n^2}$)

$$\text{MSE}(\hat{\theta}_1) = \frac{2\sigma^4}{n-1}$$

$$\text{MSE}(\hat{\theta}_2) = \frac{2\sigma^4(n-1)}{n^2} + \frac{\sigma^4}{n^2} = \frac{\sigma^4(2n-1)}{n^2}$$

For any $n \geq 2$: $\text{MSE}(\hat{\theta}_2) < \text{MSE}(\hat{\theta}_1)$! The biased MLE actually has smaller total error.

13.8 MATLAB Examples

Example 1: MLE for Exponential Distribution

MATLAB

```
% MLE for Exponential: Estimate failure rate
lambda_true = 0.005; % True failure rate
n = 30;

% Simulate data
data = exprnd(1/lambda_true, 1, n);

% MLE
lambda_hat = 1 / mean(data);
fprintf('True  $\lambda = %.4f$ , MLE  $\hat{\lambda} = %.4f$ \n', lambda_true, lambda_hat);

% Repeated experiments to see variability
N_exp = 10000;
lambda_estimates = zeros(1, N_exp);
for i = 1:N_exp
    d = exprnd(1/lambda_true, 1, n);
    lambda_estimates(i) = 1/mean(d);
end

fprintf('E[ $\hat{\lambda}$ ] = %.5f (biased: true is %.5f)\n', mean(lambda_estimates), lambda_true);
fprintf('Std( $\hat{\lambda}$ ) = %.5f\n', std(lambda_estimates));

figure;
histogram(lambda_estimates, 80, 'Normalization', 'pdf');
xlabel(' $\hat{\lambda}$ '); ylabel('PDF'); title('Sampling Distribution of MLE  $\hat{\lambda}$ ');

```

Example 2: MLE vs Method of Moments

MATLAB

```

%% Compare MLE and MoM for Uniform(0, θ)
% MLE:  $\hat{\theta} = \max(X_1, \dots, X_n)$ 
% MoM:  $\hat{\theta} = 2\bar{X}$ 
theta_true = 10; n = 20;
N_exp = 50000;

MLE_est = zeros(1, N_exp);
MoM_est = zeros(1, N_exp);

for i = 1:N_exp
    data = theta_true * rand(1, n);
    MLE_est(i) = max(data);
    MoM_est(i) = 2 * mean(data);
end

fprintf('Method      | E[ $\hat{\theta}$ ] | Bias | Var | MSE\n');
fprintf('MLE (max)    | %.3f | %.3f | %.3f | %.3f\n', ...
    mean(MLE_est), mean(MLE_est)-theta_true, var(MLE_est), ...
    var(MLE_est)+(mean(MLE_est)-theta_true)^2);
fprintf('MoM (2*mean) | %.3f | %.3f | %.3f | %.3f\n', ...
    mean(MoM_est), mean(MoM_est)-theta_true, var(MoM_est), ...
    var(MoM_est)+(mean(MoM_est)-theta_true)^2);

```

Example 3: MLE for Gaussian Parameters

MATLAB

```

%% MLE for Gaussian: Estimate both  $\mu$  and  $\sigma^2$ 
mu_true = 5; sigma2_true = 4; n = 50;
N_exp = 10000;

mu_hat = zeros(1, N_exp);
sigma2_MLE = zeros(1, N_exp);
sigma2_unbiased = zeros(1, N_exp);

for i = 1:N_exp
    data = mu_true + sqrt(sigma2_true)*randn(1, n);
    mu_hat(i) = mean(data);
    sigma2_MLE(i) = mean((data - mean(data)).^2); % MLE: divides by n
    sigma2_unbiased(i) = var(data); % Unbiased: divides by n-1
end

fprintf('μ: E=%.3f (true=%.1f), Var=%.4f\n', mean(mu_hat), mu_true, var(mu_hat));
fprintf('σ²_MLE: E=%.3f (true=%.1f, biased!)\n', mean(sigma2_MLE), sigma2_true);
fprintf('σ²_unbiased: E=%.3f (true=%.1f)\n', mean(sigma2_unbiased), sigma2_true);

```

13.9 Practice Problems

Problem 1

Data from $N(\mu, 9)$: $\bar{x} = 4.2$, $n = 25$.

- (a) Give the MLE of μ . (b) What is $\text{Var}(\hat{\mu})$? (c) Is $\bar{X}$ efficient for μ ?

Solution

(a) $\hat{\mu}_{\text{MLE}} = \bar{X} = 4.2$

(b) $\text{Var}(\bar{X}) = \frac{\sigma^2}{n} = \frac{9}{25} = 0.36$

(c) CRLB for $N(\mu, \sigma^2)$: $\frac{1}{n \cdot I(\mu)} = \frac{\sigma^2}{n} = 0.36$. $\bar{X}$ achieves CRLB $\rightarrow$ yes, efficient.

Problem 2

Component lifetimes (hours): 120, 350, 200, 480, 90, 560, 310, 175, 420, 280.

- (a) Estimate λ (failure rate) by MLE assuming $\text{Exp}(\lambda)$. (b) Estimate MTTF.

Solution

(a) $\bar{X} = \frac{120 + 350 + 200 + 480 + 90 + 560 + 310 + 175 + 420 + 280}{10} = 298.5$.

$\hat{\lambda} = \frac{1}{298.5} = 0.00335$ failures/hour.

(b) $\text{MTTF} = \frac{1}{\hat{\lambda}} = \bar{X} = 298.5$ hours.

Problem 3

For $\text{Poisson}(\lambda)$ data: $x_1, \dots, x_n$. Derive the MLE of λ .

Solution $\ell(\lambda) = \sum [x_i \ln(\lambda) - \lambda - \ln(x_i!)] = (\sum x_i) \ln(\lambda) - n\lambda - \sum \ln(x_i!)$

$$\frac{d\ell}{d\lambda} = \frac{\sum x_i}{\lambda} - n = 0 \rightarrow \hat{\lambda}_{\text{MLE}} = \frac{1}{n} \sum x_i = \bar{X}.$$

The MLE for Poisson rate is the sample mean.

Next Module: Module 14 — Confidence Intervals